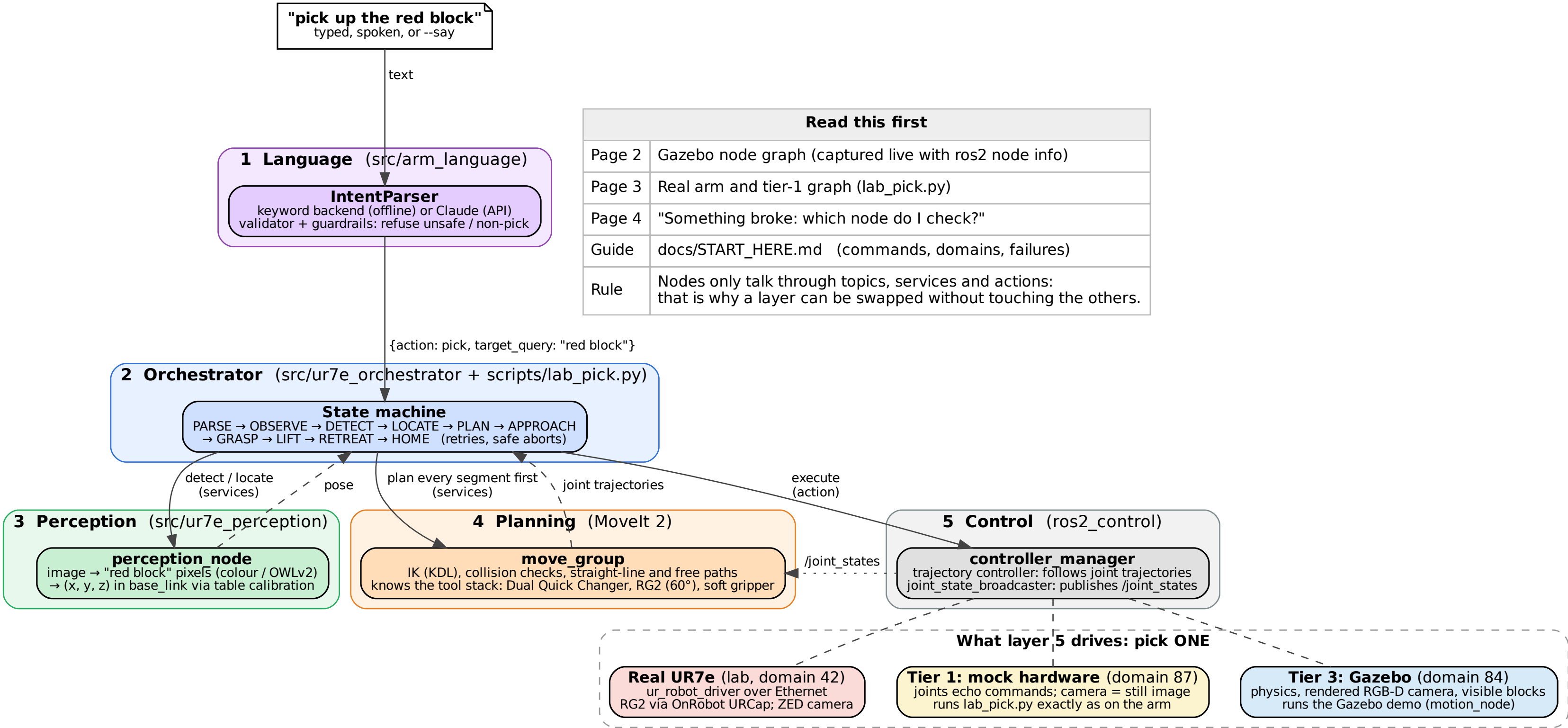


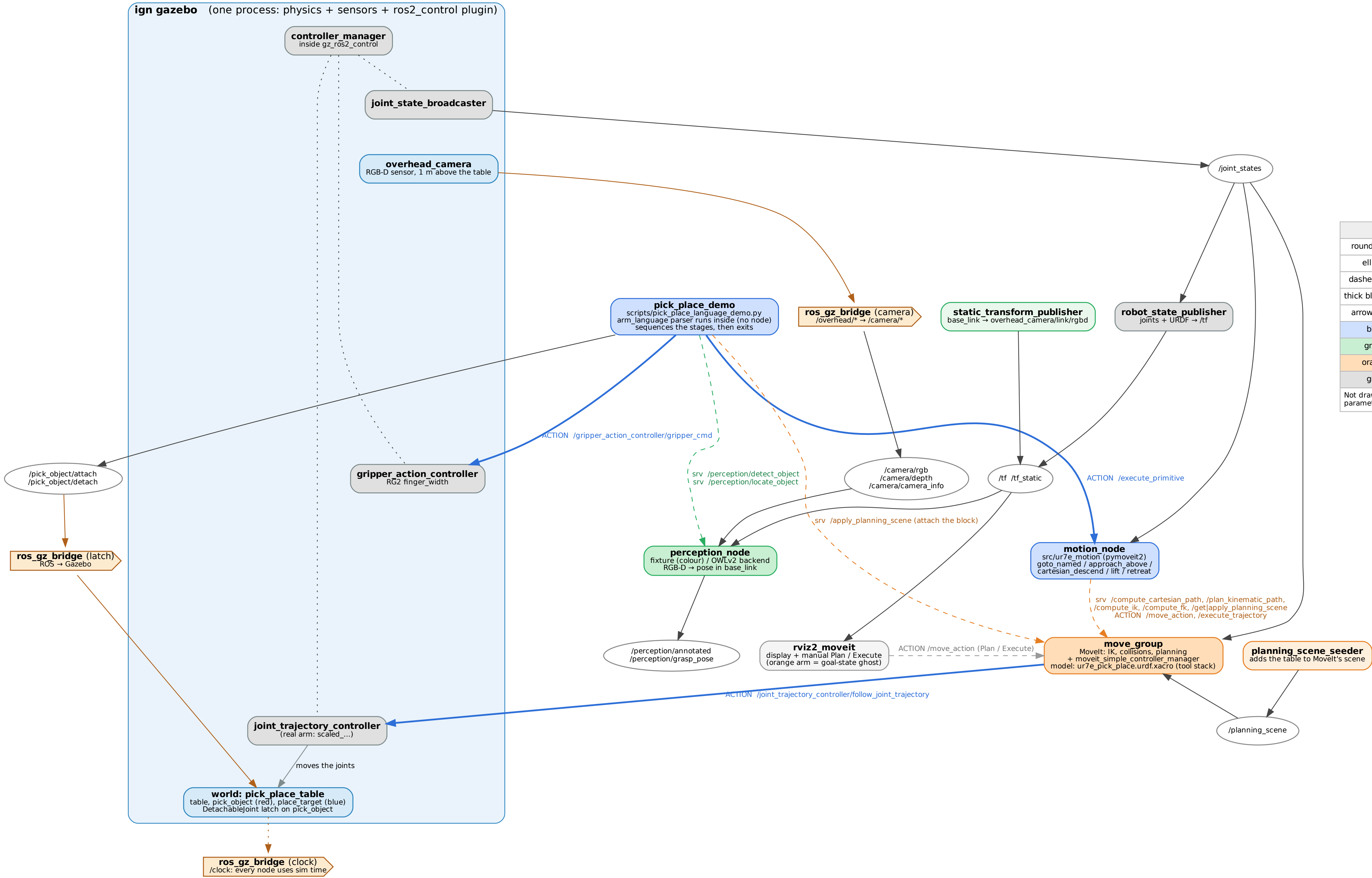
UR7e language pick-and-place: how the pieces connect

One sentence travels down five layers. Same code on the real arm and in simulation; only the bottom layer is swapped. (2026-10-08)



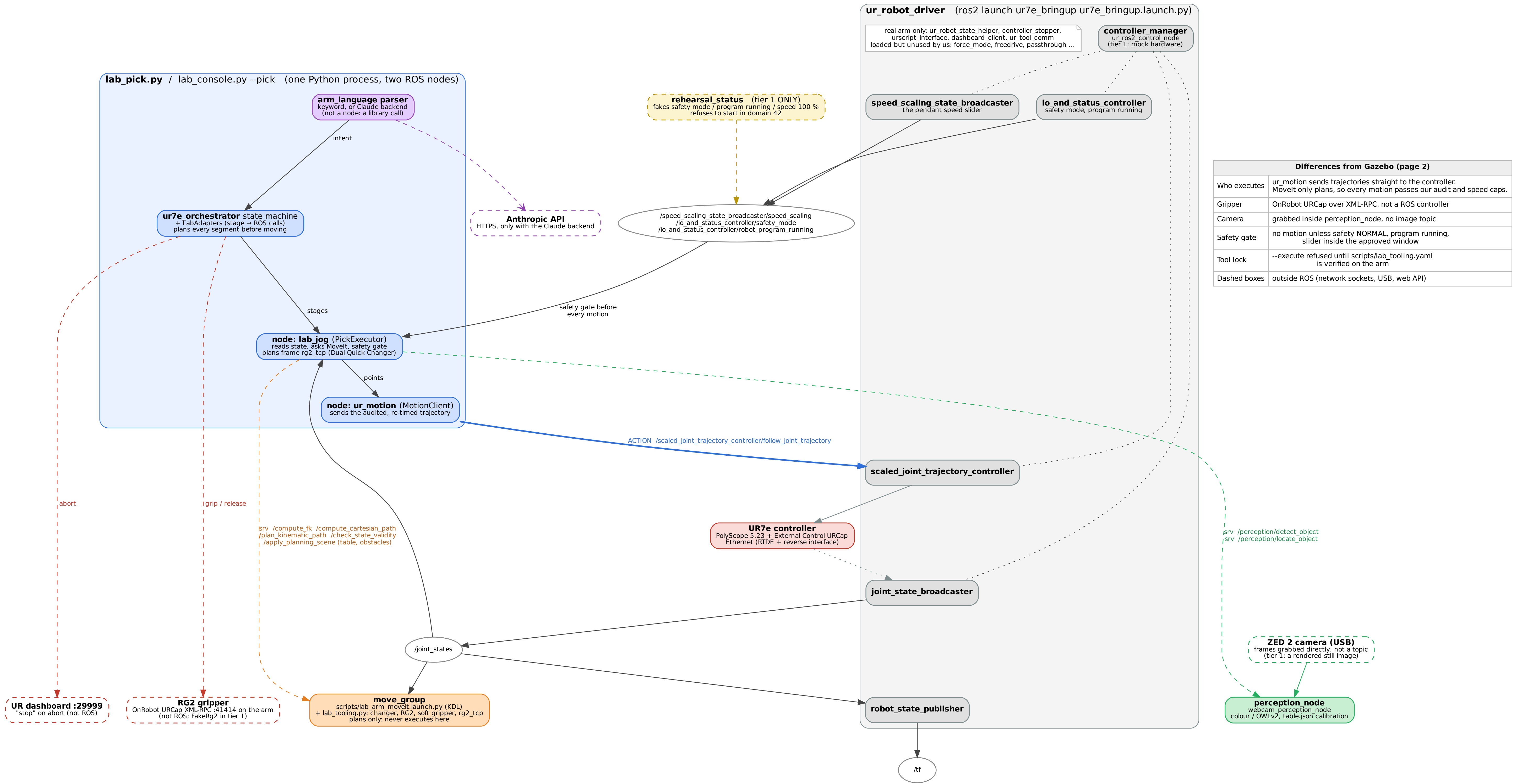
Tier 3: Gazebo node graph

ros2 launch ur7e_perception gazebo_demo.launch.py + python3 scripts/pick_place_language_demo.py "pick up the red block" (domain 84; captured live 2026-10-08)



Real UR7e (and tier 1): the lab_pick.py graph

python3 scripts/lab_pick.py --say "pick up the red block" (lab: domain 42, laptop/jetson + arm on Ethernet. Tier 1: domain 87, mock hardware, same code.)



Something broke: which node do I check?

Always first: am I in the container, on the right domain? `whoami`; `pwd`; `echo $ROS_DOMAIN_ID` (sim 87 / Gazebo 84 / lab 42)

Symptom	Node / piece to check	Command
No fresh joint state -- is the driver running?	joint_state_broadcaster (inside controller_manager) or this terminal is on another domain	<code>ros2 topic hz /joint_states</code>
Plan works, Execute does nothing / times out	the trajectory controller is not active	<code>ros2 control list_controllers</code>
PLAN unreachable / no collision-free hover	move_group: the target, the scene, or a misplaced block	look in RViz; Gazebo: reset pick_object (START_HERE 5b)
/perception/detect_object unavailable	perception_node not up (or crashed on start)	<code>ros2 service list grep perception</code>
not_found for an object that is there	perception_node backend / colour words / reach region	open the snapshot dir; /perception/annotated
execution gate closed / safety_mode_not_normal	io_and_status_controller + speed_scaling (lab) rehearsal_status (tier 1)	<code>ros2 topic echo /io_and_status_controller/safety_mode</code>
Failed to find a free participant index	not a node: CYCLONEDDS_URI missing in this terminal	the full export line (START_HERE 4)
lab_tooling.yaml is not verified	not a node: the Dual Quick Changer check	docs/vendor/cad/README.md; sim: --allow-unverified-tooling
RViz shows no arm / wrong pose	robot_state_publisher (/tf)	<code>ros2 run tf2_ros tf2_echo base_link tool0</code>
Gazebo: block not lifted after a "successful" pick	latch bridge (/pick_object/attach)	Entity Tree → pick_object → Pose z (0.14 = lifted)

See the graph yourself (live)	
<code>ros2 node list</code>	who is running
<code>ros2 node info /move_group</code>	a node's topics, services, actions (how page 2 was made)
<code>ros2 topic list</code> / <code>ros2 topic echo /joint_states --once</code>	streams and one sample
<code>ros2 service list</code> / <code>ros2 action list</code>	request/reply and long-running goals
<code>rqt_graph</code>	this picture, auto-drawn (topics only)
Words: node = one process doing one job. topic = a stream anyone can read (/joint_states). service = ask once, get one answer (/compute_ik). action = a long job with progress and a result (follow_joint_trajectory). domain = which nodes can find each other at all. Sources of these pages: docs/diagrams/*.dot (run build.sh).	